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***Abstract*—This document is a model and instructions for LaTeX. This and the IEEEtran.cls file define the components of your paper [title, text, heads, etc.]. *CRITICAL: Do Not Use Symbols, Special Characters, Footnotes, or Math in Paper Title or Abstract.**

***Index Terms*—component, formatting, style, styling, insert.**

I. Introduction

This document is a model and instructions for LaTeX. Please observe the conference page limits. For more information about how to become an IEEE Conference author or how to write your paper, please visit IEEE Conference Author Center website: https://conferences.ieeeauthorcenter.ieee.org/.

II. System Model

A. Semantic Encoder

The source image is encoded into two distinct semantic features, namely the bounding box coordinate of the target person and the cropped image of the target person from the source image, utilizing two semantic extractors based on the YOLOv8 model and down sampling. The bounding box coordinate is extracted only using YOLOv8 model, the cropped image is extracted using YOLOv8 model with down sampling for further compression. For simplicity, we use subscripts 1 and 2 to replace subscripts coordinate and cropped image of person, respectively, in the following discussion. The ith extracted feature can be expressed by

$$\mathbf{S}_i = \mathcal{E}_i\left(\mathbf{X} \mid \boldsymbol{\theta}_i^*\right), \forall i \in \{1,2\}. \tag{1}$$

where $\mathcal{E}_i\left(\mathbf{X} \mid \boldsymbol{\theta}_i^*\right)$ is the ith model with $\boldsymbol{\theta}_i^*$ being network parameters.

To enhance domain adaptability, transfer learning [1] is employed by fine-tuning the pre-trained YOLOv8 weights using our wilderness search-and-rescue personnel dataset. This fine-tuning process enables the model to align more effectively with the semantic extraction task, thereby improving its robustness and localization accuracy in complex wilderness scenarios.

As reported in Table X, the YOLOv8 model fine-tuned on the wilderness search-and-rescue dataset shows a substantial improvement in detection accuracy, compared with the model using pre-trained weights without fine-tuning, thereby validating the effectiveness of the proposed transfer learning strategy.

To align with existing digital communication systems, the semantic feature $\mathbf{S}_i$ is transformed into a bit sequence $\mathbf{K}_i$, which can be expressed as

$$\mathbf{K}_i = \mathcal{B}\left(\mathbf{S}_i\right), \tag{2}$$

where $\mathcal{B}\left(\cdot\right)$ is a binary mapping function.

B. Transmission Module

Due to the different importance of the semantic data streams, multi-stream transmissions are considered in our system. The received data streams are expressed as

$$\left[\hat{\mathbf{K}}_1, \hat{\mathbf{K}}_2\right] = \mathcal{T}\left(\left[\mathbf{K}_1, \mathbf{K}_2\right]\right), \tag{3}$$

where $\mathcal{T}\left(\cdot\right)$ is the transmission scheme, which comprises diversity/combining, the channel coding/decoding and modulation/demodulation components.

In the transmission scheme, the bit sequences $\mathbf{K}_1$ and $\mathbf{K}_2$ are all encoded using LDPC channel coding, but with different code rates and modulation schemes. The bit sequence $\mathbf{K}_1$ is transmitted using low-order modulation, low rate channel coding, and a diversity scheme, whereas $\mathbf{K}_2$ is conveyed via an adaptive coded modulation scheme.

An additive white Gaussian noise channel combined with Rayleigh fading is considered as the channel model in this paper to realistically simulate the wireless transmission environment in wilderness search-and-rescue scenarios [1].

C. Semantic Decoder

In the semantic decoder, the received semantic data streams $\hat{\mathbf{K}}_i$ are first reconverted to the semantic features $\hat{\mathbf{S}}_i$, which can be expressed by

$$\hat{\mathbf{S}}_i = \mathcal{B}^{-1}\left(\hat{\mathbf{K}}_i\right), \tag{4}$$

where $\mathcal{B}^{-1}\left(\cdot\right)$ is the inverse operation of $\mathcal{B}\left(\cdot\right)$. After obtaining the semantic features $\hat{\mathbf{S}}_i$, a background image $\mathbf{U}$ is required for further processing. The background image is provided from the knowledge base generated by stable diffusion inpainting model. Finally, the semantic features $\hat{\mathbf{S}}_i$ and the background image $\mathbf{U}$ are forwarded to the synthesizer D to synthesize $\hat{\mathbf{X}}$, which can be expressed as

$$\hat{\mathbf{X}} = D\left(\hat{\mathbf{S}}_1, \hat{\mathbf{S}}_2, \mathbf{U}\right) = \mathcal{D}\left(\hat{\mathbf{K}}_1, \hat{\mathbf{K}}_2, \mathbf{U}\right), \tag{5}$$

where $\mathcal{D}$ is the semantic decoder.

TABLE I
CODING AND MODULATION SCHEMES SELECTED

	Coding Rate	Modulation Mode	R_m (bits/sym)
CMS1	$4/5$	BPSK	4/5
CMS2	$2/3$	BPSK	2/3
CMS3	$1/4$	16QAM	1
CMS4	$1/8$	16QAM	1/2
CMS5	$1/8$	64QAM	3/4
CMS6	$1/16$	64QAM	3/8

III. THE PROPOSED SEMANTIC COMMUNICATION SCHEME

A. Adaptive Coding and Modulation

In semantic communication systems, transmitted signals over noisy channels are still affected by various channel impairments, including path loss, shadowing, fading, and noise, which deteriorate the quality of the received signal. To overcome these impairments and enhance reliability while increasing system capacity, link adaptation techniques are employed to modify the transmitted signal before transmission according to the current channel state. One of the most widely used approaches is Adaptive Coding and Modulation (ACM). In ACM, the modulation order and coding rate are unfixed. Lower-order modulation schemes and lower coding rates are selected when the channel is degraded, whereas higher-order modulation schemes and higher coding rates are used under favorable channel conditions to maximize the transmission rate.

In this paper, the coordinates carry more critical semantic information. The reliability of the coordinates are more important than the cropped person image for the semantic decoder at the receiver. Therefore, the semantic data streams $\mathbf{K}_1$ can be transmitted using low-order modulation and low code rate, specifically 1/4 code rate and BPSK modulation. For the semantic data streams $\mathbf{K}_1$, we employ BPSK, 16-QAM, and 64-QAM modulations in combination with different code rates of LDPC coding. The code rates used are 4/5, 2/3, 1/4, 1/8, and 1/16. An adaptive coded modulation scheme used in this system is shown as Table I.

For the nth coded modulation scheme CMSn, if the code rate is R and M denotes the modulation order, the number of information bits carried by each symbol is $\mathrm{R_n} = \mathrm{R} \cdot \log_2 \mathrm{M}$.

B. Spatial Diversity and Combining

Define abbreviations and acronyms the first time they are used in the text.

C. Background Image Knowledge Base

With the advancement of knowledge engineering, an increasing number of knowledge-based systems have been developed and used in various domains.

The proposed background image knowledge base is applied to the reconstruction of target person images in our wilderness search-and-rescue task. The background image knowledge base provide a background image as input to the synthesizer.

Inpainting is the task of filling masked regions of an image with new content either because parts of the image are corrupted or to replace existing but undesired content within the image [**?**].

Stable Diffusion Inpainting is a latent text-to-image diffusion model that can generate photo-realistic images from text input and can also perform inpainting by using a mask.

In this paper, the Stable Diffusion Inpainting is used to generate a background image $\mathbf{U}$ background image knowledge base. To improve its performance, several improvements are applied to the model inputs. The Stable Diffusion Inpainting model requires three inputs, including a text prompt, an input image, and a corresponding mask image. The mask image is obtained by directly processing the input image with a pretrained background removal model, namely rembg.

D. Units

- Use either SI (MKS) or CGS as primary units. (SI units are encouraged.) English units may be used as secondary units (in parentheses). An exception would be the use of English units as identifiers in trade, such as "3.5-inch disk drive".
- Avoid combining SI and CGS units, such as current in amperes and magnetic field in oersteds. This often leads to confusion because equations do not balance dimensionally. If you must use mixed units, clearly state the units for each quantity that you use in an equation.
- Do not mix complete spellings and abbreviations of units: "Wb/m^2" or "webers per square meter", not "webers/m^2". Spell out units when they appear in text: ". . . a few henries", not ". . . a few H".
- Use a zero before decimal points: "0.25", not ".25". Use "cm^3", not "cc".)

E. Equations

Number equations consecutively. To make your equations more compact, you may use the solidus (/), the exp function, or appropriate exponents. Italicize Roman symbols for quantities and variables, but not Greek symbols. Use a long dash rather than a hyphen for a minus sign. Punctuate equations with commas or periods when they are part of a sentence, as in:

$$a + b = \gamma \tag{6}$$

Be sure that the symbols in your equation have been defined before or immediately following the equation. Use "(6)", not "Eq. (6)" or "equation (6)", except at the beginning of a sentence: "Equation (6) is . . ."

F. LaTeX-Specific Advice

Please use "soft" (e.g., `\eqref{Eq}`) cross references instead of "hard" references (e.g., `(1)`). That will make it possible to combine sections, add equations, or change the order of figures or citations without having to go through the file line by line.

Please don't use the `{eqnarray}` equation environment. Use `{align}` or `{IEEEeqnarray}` instead. The `{eqnarray}` environment leaves unsightly spaces around relation symbols.

Please note that the `{subequations}` environment in LaTeX will increment the main equation counter even when there are no equation numbers displayed. If you forget that, you might write an article in which the equation numbers skip from (17) to (20), causing the copy editors to wonder if you've discovered a new method of counting.

BibTeX does not work by magic. It doesn't get the bibliographic data from thin air but from .bib files. If you use BibTeX to produce a bibliography you must send the .bib files.

LaTeX can't read your mind. If you assign the same label to a subsubsection and a table, you might find that Table I has been cross referenced as Table IV-B3.

LaTeX does not have precognitive abilities. If you put a `\label` command before the command that updates the counter it's supposed to be using, the label will pick up the last counter to be cross referenced instead. In particular, a `\label` command should not go before the caption of a figure or a table.

Do not use `\nonumber` inside the `{array}` environment. It will not stop equation numbers inside `{array}` (there won't be any anyway) and it might stop a wanted equation number in the surrounding equation.

G. Some Common Mistakes

- The word "data" is plural, not singular.
- The subscript for the permeability of vacuum μ_0, and other common scientific constants, is zero with subscript formatting, not a lowercase letter "o".
- In American English, commas, semicolons, periods, question and exclamation marks are located within quotation marks only when a complete thought or name is cited, such as a title or full quotation. When quotation marks are used, instead of a bold or italic typeface, to highlight a word or phrase, punctuation should appear outside of the quotation marks. A parenthetical phrase or statement at the end of a sentence is punctuated outside of the closing parenthesis (like this). (A parenthetical sentence is punctuated within the parentheses.)
- A graph within a graph is an "inset", not an "insert". The word alternatively is preferred to the word "alternately" (unless you really mean something that alternates).
- Do not use the word "essentially" to mean "approximately" or "effectively".
- In your paper title, if the words "that uses" can accurately replace the word "using", capitalize the "u"; if not, keep using lower-cased.
- Be aware of the different meanings of the homophones "affect" and "effect", "complement" and "compliment", "discreet" and "discrete", "principal" and "principle".
- Do not confuse "imply" and "infer".
- The prefix "non" is not a word; it should be joined to the word it modifies, usually without a hyphen.
- There is no period after the "et" in the Latin abbreviation "et al.".
- The abbreviation "i.e." means "that is", and the abbreviation "e.g." means "for example".

An excellent style manual for science writers is [**?**].

H. Authors and Affiliations

The class file is designed for, but not limited to, six authors. A minimum of one author is required for all conference articles. Author names should be listed starting from left to right and then moving down to the next line. This is the author sequence that will be used in future citations and by indexing services. Names should not be listed in columns nor group by affiliation. Please keep your affiliations as succinct as possible (for example, do not differentiate among departments of the same organization).

I. Identify the Headings

Headings, or heads, are organizational devices that guide the reader through your paper. There are two types: component heads and text heads.

Component heads identify the different components of your paper and are not topically subordinate to each other. Examples include Acknowledgments and References and, for these, the correct style to use is "Heading 5". Use "figure caption" for your Figure captions, and "table head" for your table title. Run-in heads, such as "Abstract", will require you to apply a style (in this case, italic) in addition to the style provided by the drop down menu to differentiate the head from the text.

Text heads organize the topics on a relational, hierarchical basis. For example, the paper title is the primary text head because all subsequent material relates and elaborates on this one topic. If there are two or more sub-topics, the next level head (uppercase Roman numerals) should be used and, conversely, if there are not at least two sub-topics, then no subheads should be introduced.

J. Figures and Tables

a) Positioning Figures and Tables: Place figures and tables at the top and bottom of columns. Avoid placing them in the middle of columns. Large figures and tables may span across both columns. Figure captions should be below the figures; table heads should appear above the tables. Insert figures and tables after they are cited in the text. Use the abbreviation "Fig. 1", even at the beginning of a sentence.

Figure Labels: Use 8 point Times New Roman for Figure labels. Use words rather than symbols or abbreviations when writing Figure axis labels to avoid confusing the reader. As an example, write the quantity "Magnetization", or "Magnetization, M", not just "M". If including units in the label, present them within parentheses. Do not label axes only with units. In the example, write "Magnetization (A/m)" or "Magnetization {A[m(1)]}", not just "A/m". Do not label axes with a ratio of quantities and units. For example, write "Temperature (K)", not "Temperature/K".

Acknowledgment

The preferred spelling of the word "acknowledgment" in America is without an "e" after the "g". Avoid the stilted

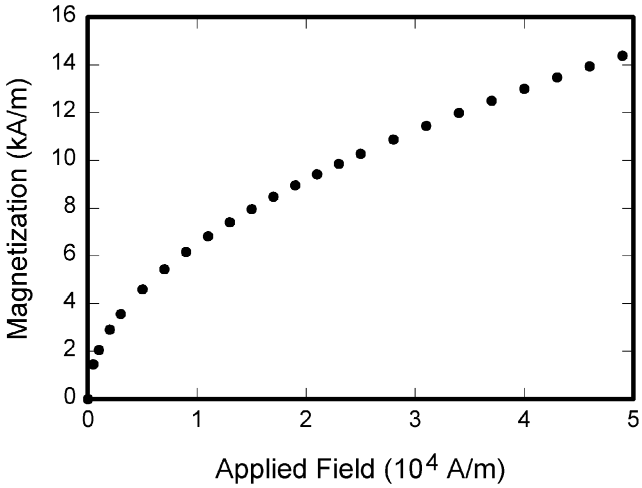

Fig. 1. Example of a figure caption.

expression "one of us (R. B. G.) thanks ...". Instead, try "R. B. G. thanks...". Put sponsor acknowledgments in the unnumbered footnote on the first page.

REFERENCES

Please number citations consecutively within brackets [**?**]. The sentence punctuation follows the bracket [**?**]. Refer simply to the reference number, as in [**?**]—do not use "Ref. [**?**]" or "reference [**?**]" except at the beginning of a sentence: "Reference [**?**] was the first ..."

Number footnotes separately in superscripts. Place the actual footnote at the bottom of the column in which it was cited. Do not put footnotes in the abstract or reference list. Use letters for table footnotes.

Unless there are six authors or more give all authors' names; do not use "et al.". Papers that have not been published, even if they have been submitted for publication, should be cited as "unpublished" [**?**]. Papers that have been accepted for publication should be cited as "in press" [**?**]. Capitalize only the first word in a paper title, except for proper nouns and element symbols.

For papers published in translation journals, please give the English citation first, followed by the original foreign-language citation.

REFERENCES

[1] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 10 684–10 695.